

Effectiveness of an immediate postoperative electroacupuncture session for the recovery of gastrointestinal function after laparoscopic cholecystectomy: a randomized controlled trial

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Abstract

Objective: The objective of the study is to evaluate the effectiveness and safety of one session of electroacupuncture (EA) in the immediate postoperative period for the recovery of gastrointestinal function after laparoscopic cholecystectomy (LC).

Methods: A randomized, assessor-blinded controlled trial was conducted from October 2022 to March 2023 with 1:1 allocation to the intervention group and the control group. Eighty-four patients with benign gallbladder disease based on conventional diagnostic criteria were randomly assigned to one of the two different groups. The intervention group received a 30 min EA treatment at LI4, PC6, ST36 and LR3 (2/100 Hz frequency, 1 mA average intensity) that was given immediately after surgery. All patients received standard perioperative management based on an enhanced recovery after surgery (ERAS) protocol. Time to first flatus was the primary outcome. Secondary outcomes included time to first defecation and first normal bowel sound, as well as evaluation of abdominal distension/pain and postoperative nausea and vomiting (PONV).

Results: The intervention group had a shorter time to first flatus and time to first normal bowel sound. They also had a lower overall incidence of postoperative abdominal distension, pain and PONV over the first 24 h. The degree/grading of abdominal distension was also lower at 12 h and that of abdominal pain and PONV was lower at both 12 and 24 h postoperatively.

Conclusion: One session of EA immediately after surgery may reduce the incidence and severity of postoperative gastrointestinal dysfunction (POGD).

Trial registration number: ChiCTR2200064748 (Chinese Clinical Trial Registry)

Keywords

electroacupuncture, enhanced recovery after surgery, gastrointestinal function, laparoscopic cholecystectomy, randomized controlled trial

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Introduction

In the realm of postoperative care, postoperative gastrointestinal dysfunction (POGD) represents a constellation of signs and symptoms that arise as a result of perioperative factors impacting gastrointestinal function.¹ These may include reduced or absent bowel sounds, poor appetite, nausea, abdominal pain, abdominal distension, poor gastric motility and, in severe cases, intestinal obstruction.² Clinical studies have shown that the incidence of POGD after abdominal gastrointestinal surgery can exceed 40%,^{3–5} while the incidence after biliary surgery may be >50%.⁶ In recent years, there has been a marked increase in the incidence of benign gallbladder diseases, including gallstones, acute and chronic cholecystitis, simple gallbladder polyps and gallbladder adenomyomatosis.^{7–10} Currently, China lacks national epidemiological data on benign gallbladder diseases; however, large-scale regional surveys suggest that the overall incidence of these conditions may exceed 20%.^{10–13}

Enhanced recovery after surgery (ERAS) refers to a series of effective treatment measures taken during the perioperative period with the goal of accelerating the speed of postoperative recovery and reducing the incidence of postoperative complications, all while prioritizing patient-centered care and adopting the principles of evidence-based medicine.¹⁴ The concept of ERAS was introduced to China by academician Li Jieshou in 2006.¹⁵ During the process of local integration of ERAS in China over more than a decade, the incorporation of traditional Chinese medicine (TCM)-related techniques, which arguably have some unique advantages, has been increasing. In 2017, Professor Huang Jianping from the Affiliated Shuguang Hospital of Shanghai University of Traditional Chinese Medicine first proposed the concept of perioperative Chinese medicine enhanced recovery after surgery (CMERAS) with distinctive Chinese characteristics.¹⁶ He also pointed out that, in contrast to the concept of TCM rehabilitation, CMERAS aims to reduce traumatic stress and accelerate postoperative recovery, focusing on solving key issues during the perioperative period. It utilizes an integrative approach combining both TCM and Western medicine to provide what we consider to be the best perioperative diagnostic and treatment pathway for patients.¹⁷ As the ERAS approach has continued to evolve and improve, acupuncture has emerged as a complementary and/or alternative therapy for rapid surgical recovery due to its relative safety, convenience and low rate of adverse reactions, thus playing a unique role in surgical recovery.^{18–20}

In this study, we adopted the concept of CMERAS, which involves intervening with electroacupuncture (EA) immediately after surgery, and observed the recovery of gastrointestinal function in patients after laparoscopic cholecystectomy (LC). Our recent meta-analysis indicated that EA is effective at promoting gastrointestinal recovery

following LC.²¹ However, the seven trials included in that meta-analysis had a high risk of bias in certain domains, including lack of blinding and allocation concealment, which limits the reliability of the pooled clinical evidence. Accordingly, we aimed to conduct a high-quality randomized controlled trial (RCT) to further assess the effectiveness and safety of acupuncture as a potential adjunct to standard perioperative care in this patient population. We anticipated that our findings would offer further insight into the role of a standardized application of acupuncture to potentially aid the recovery of gastrointestinal function after surgery.

Methods

Study design

A single-blind parallel randomized controlled trial (RCT) was conducted. Eighty-four participants were randomly divided into an intervention group and a control group in a 1:1 allocation ratio. The protocol for this RCT was designed in accordance with the Consolidated Standards of Reporting Trials (CONSORT) and the STAndards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA) guidelines.^{22,23} The intervention group received a 30 min EA treatment immediately after surgery. All patients received standard perioperative management based on the ERAS protocol. This study was approved by the Ethics Committee of Shanghai University of Traditional Chinese Medicine Affiliated Shuguang Hospital (2021-1017-92-01) and registered in the Chinese Clinical Trial Registry (ChiCTR2200064748) on 15 October 2022, prior to the recruitment of the first participant.

Study participants

Patients who were hospitalized in the Department of Hepatobiliary Surgery at Shuguang Hospital Affiliated to Shanghai University of Traditional Chinese Medicine from 31 October 2022 to 31 March 2023 and needed LC were enrolled for this study. The diagnostic criteria were determined according to established indications for LC.²⁴ To meet the inclusion criteria, patients needed to: (a) be between 18 and 70 years old; (b) have an American Society of Anesthesiologists (ASA) anesthesia risk grade of I–II; (c) have no symptoms of gastrointestinal dysfunction before surgery and no evidence of sepsis, bacteremia or obvious abdominal infection; and (d) be willing to sign the informed consent form.

Exclusion criteria were determined as follows: (a) pregnancy or lactation; (b) significant organ dysfunction or systemic disease; (c) psychiatric disorders or language barriers; (d) allergies or contraindications to anesthesia; (e) participation in another clinical trial in the preceding 4 weeks; and (f) history of a prior surgical procedure.

Blinking

This study employed a single-blind design. In line with the principles of blinding, strict separation of roles was maintained among the research team. Surgeons, anesthetists and outcome assessors were blinded to group allocation. Only the acupuncturists, who were not blinded, were aware of the group assignments; however, they were instructed not to disclose this information to any other researchers or participants. In addition, all study personnel received training on specific procedures and were required to strictly follow the principle of blinding. The participants received the acupuncture intervention shortly after the LC surgery, at a point in time at which they had not regained full consciousness. Thus, participant blinding was a possibility but could not be assured, given the variable time interval between the end of general anesthesia and the return of awareness.

Sample size calculation

The target sample size was calculated using the primary outcome measure, which was time to first flatus, using the equation below. In a prior study,²⁵ the time to first flatus after surgery in the intervention group was 54.15 ± 11.27 h, while that of the routine ERAS group was 60.89 ± 12.31 h. Based on this variability, it was estimated that 76 patients would be needed to detect this effect size at 80% power with parallel groups, an allocation ratio of 1:1 and $\alpha=0.05$. Considering a predicted dropout rate of 10%, the target sample size was inflated to 84 cases ($n=42$ patients in each group):

$$N = \left[\frac{(U_{\alpha} + U_{\beta}) \delta}{\mu_1 - \mu_2} \right]^2 \times \left(\frac{1}{k} + \frac{1}{1-k} \right)$$

$$N = n_1 + n_2$$

Randomization and allocation

Block randomization was used to divide the 84 subjects into two groups using the Statistical Package for the Social Sciences (SPSS) version 23.0 (IBM Corp., Armonk, NY, USA). The randomization results were printed on cards, which were then placed into sealed envelopes made of parchment paper and sequentially numbered. According to the order in which the subjects were enrolled, the corresponding envelopes were opened, and the corresponding treatment methods were applied.

Intervention

Based on the *Guidelines for Clinical Practice in Enhanced Recovery after Surgery (ERAS) in China*,²⁶ released in 2021, participants in both groups received standard perioperative management. The ERAS protocol included preoperative counseling, preoperative nutrition, avoidance of

perioperative fasting, carbohydrate loading up to 2 h before surgery, standardized anesthetic and analgesic regimens, and early mobilization. Both groups also received a standardized dose of intravenous metoclopramide (10 mg).

The intervention group received one session of EA in addition to standard care, which was administered by acupuncturists with at least 5 years of clinical experience. EA was performed immediately after surgery, starting when the patient was transferred from the operating table to a recovery bed. The patient was placed in a supine position and the skin overlying the planned needling sites was disinfected with alcohol. Sterile disposable steel needles (0.25 mm diameter $\times$ 40 mm length; Jiajian Medical Equipment Co., Ltd, Wu Xi, Jiang Su Province, China) were inserted perpendicularly to a depth of approximately 10 mm at bilateral LI4 (*Hegu*), PC6 (*Neiguan*) and LR3 (*Taichong*) and to a depth of about 15 mm at bilateral ST36 (*Zusanli*), based on clinical experience and a recent literature review.²⁷ Even reinforcing and reducing manipulation techniques were used upon achieving *de qi* sensation, defined as when the operator perceived a sinking or taut sensation in the needle. Positive and negative electrodes were connected to the needles at ST36/LR3 and LI4/PC6 on each side of the body and connected to an EA apparatus (CMNS6-2 type, Jiajian Medical Equipment Co., Ltd. Wu Xi, Jiang Su Province, China) that was used to deliver sparse-dense waves of 2/100 Hz (alternating every 1.5 s). Based on prior clinical experience, slight muscle twitching of the area surrounding the needle insertion site was used as a reference for a suitable level of stimulation (average current intensity of 1 mA). The needle retention time was 30 min.

Primary outcome

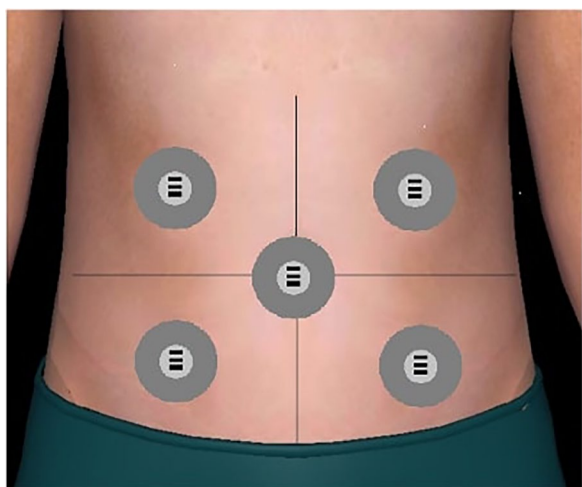
Time to first flatus. The time interval from the end of surgery (when participants were transferred off the operating table) to first flatus was recorded on trial data collection forms by LH within 48 h. LH did not have any direct contact with the participants or the acupuncturists until the whole trial was finished.

Secondary outcomes

Time to first defecation. The time interval from the end of the surgery to the first spontaneous defecation was recorded as an *a priori* secondary outcome.

Time to first bowel sound. Bowel sounds were assessed hourly in five auscultation areas (right upper, right lower, left upper, left lower and periumbilical) using a stethoscope for 1 min per area (Figure 1), starting immediately after the EA intervention and continuing until normal bowel sounds were detected. The number of hours elapsed at the onset of bowel sounds was recorded as an *a priori* secondary outcome.

Figure 1. Five auscultation areas.



Abdominal distension. This *post hoc* secondary outcome was assessed and recorded at 12 and 24 h post-surgery using a four-level grading system:

- Grade 0: no sense of abdominal distension (0 points)
- Grade I: mild abdominal distension with a sensation of gas movement in the abdomen and no distending pain at the incision site (1 point)
- Grade II: moderate abdominal distension with abdominal swelling and distending pain at the incision site that was tolerable (2 points)
- Grade III: severe abdominal distension with restlessness, groaning and even difficulty breathing in addition to symptoms consistent with grade II (3 points)

Abdominal pain. This *post hoc* secondary outcome was assessed and recorded at 12 and 24 h post-surgery using the Prince-Henry standard, which classifies pain into five levels:

- Grade 0: absent of pain when coughing (0 points)
- Grade I: pain when coughing (1 point)
- Grade II: pain on deep breathing, but no pain at rest (2 points)
- Grade III: pain at rest, but tolerable (3 points)
- Grade IV: severe pain at rest, intolerable (4 points)

Incidence of postoperative nausea and vomiting. This *post hoc* secondary outcome was recorded at 12 and 24 h post-surgery based on established standards.^{28,29} The assessment of postoperative nausea and vomiting (PONV) was divided into five levels:

- Grade 0: no nausea or vomiting (0 points)
- Grade I: mild nausea without vomiting (1 point)
- Grade II: transient nausea and vomiting (2 points)

- Grade III: nausea and vomiting symptoms requiring treatment (3 points)
- Grade IV: uncontrollable nausea and vomiting (4 points)

Statistical analysis

The data were analyzed using the Statistical Package for the Social Sciences (SPSS) 23.0 (SPSS Inc., Chicago, IL, USA). Categorical data were analyzed using the Chi-square test. Continuous variables with a normal distribution were presented as mean $\pm$ standard deviation and compared between groups using Student's *t* test. Non-normally distributed data were expressed as median [interquartile range] and analyzed using the Mann-Whitney U test. All tests used in this study were two-sided. $P < 0.05$ was considered statistically significant.

Results

Demographic and baseline characteristics

The flow diagram of this study is depicted in Figure 2. A total of 84 enrolled patients were randomly divided into the intervention and control groups, each comprising 42 cases. All participants exhibited good compliance and completed the trial, with no exclusions or dropouts. The baseline demographic data and clinical characteristics of the two groups were comparable with no significant differences ($p > 0.05$; Table 1).

Time to first flatus, first defecation and first bowel sound

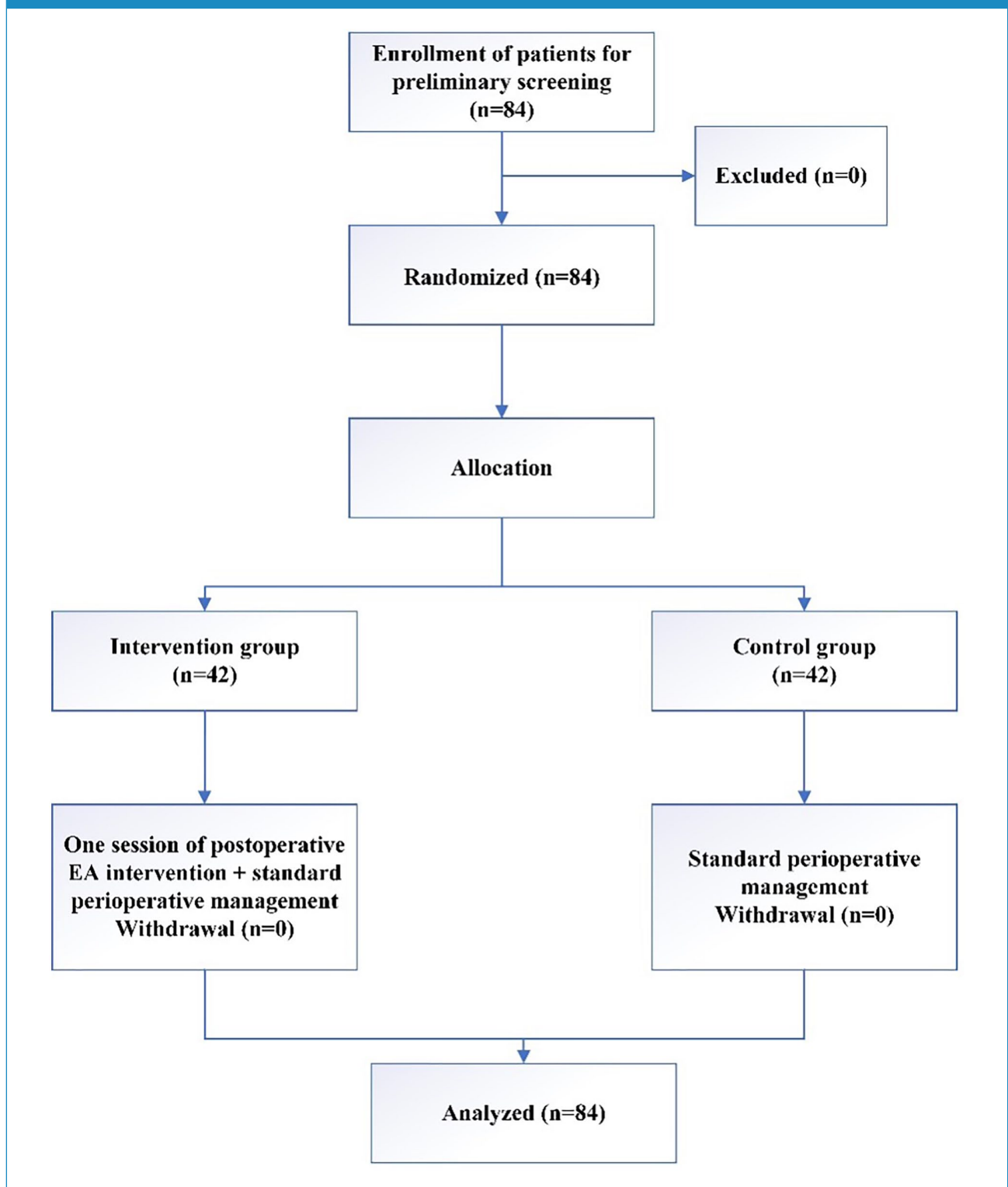
As shown in Table 2, the intervention group had a shorter time to first flatus (primary outcome measure) as well as time to first bowel sound (secondary outcome measure) compared to the control group ($p < 0.05$ and $p < 0.001$, respectively). There was no significant difference between the two groups in the time to first defecation ($p > 0.05$).

Postoperative abdominal distension

As shown in Table 3, the intervention group displayed significantly less abdominal distension at 12 h after surgery than the control group ($p < 0.01$) and tended to be less distended at 24 h, although formal statistical significance was lost by this stage ($p > 0.05$ but < 0.1). The overall incidence of postoperative abdominal distension within the first 24 h was also lower in the intervention group compared to the control group ($\chi^2 = 11.9$; $p = 0.018$).

Postoperative abdominal pain

The intervention group exhibited significantly lower abdominal pain scores than the control group at both 12 h ($p < 0.01$) and 24 h ($p < 0.05$) after LC, suggesting that

Figure 2. Flow diagram.

EA effectively reduces postoperative abdominal pain severity. The overall incidence of postoperative abdominal pain within the first 24 h was also lower in the intervention group compared to the control group ($\chi^2 = 111.2$; $p < 0.001$).

Postoperative nausea and vomiting

The incidence of PONV in the intervention group was significantly lower than that in the control group at 12

Table 1. Demographics and baseline characteristics.

Variable	Intervention group (n=42)	Control group (n=42)	p value
Gender			
- Male	17 (40.5)	12 (28.6)	0.359 ^a
- Female	25 (59.5)	30 (71.4)	
Age (years)	49.5 ± 11.83	50.6 ± 11.39	0.653 ^b
Height (cm)	164.6 ± 8.29	162.2 ± 8.33	0.206 ^b
Weight (kg)	65.1 ± 12.89	63.2 ± 11.45	0.493 ^b
Body mass index (kg/m ²)	23.8 ± 3.08	23.9 ± 3.12	0.923 ^b
GSRS score (points)	30.5 [11.25]	27.5 [10.75]	0.367 ^c
Preoperative heart rate (bpm)	69.2 ± 7.68	72.9 ± 9.26	0.053 ^b
Preoperative blood pressure (mmHg)			
- Systolic	132.8 ± 14.09	133.6 ± 17.48	0.811 ^b
- Diastolic	76.9 ± 10.05	75.4 ± 8.77	0.454 ^b
Surgical duration (min)	52.4 ± 21.86	60.1 ± 22.87	0.115 ^b

^aChi-square test (χ^2).^bUnpaired t test.^cMann-Whitney U test.

GSRS=clinical rating scale for gastrointestinal symptoms.

Data are mean ± SD, median [interquartile range] or n (%).

Table 2. Time to first flatus, first defecation and first bowel sound.

	Intervention group (n=42)	Control group (n=42)	p value
Time to first flatus (h)	25 [19, 36.25]	37 [21.75, 46.25]	0.036 ^a
Time to first defecation (h)	69.2 ± 27.01	74.8 ± 29.88	0.371 ^b
Time to first normal bowel sound (h)	1 [1, 2]	4 [3, 5]	<0.001 ^a

^aMann-Whitney U test.^bUnpaired t test.

Data are mean ± SD or median [interquartile range].

and 24 h (both $p < 0.05$) after surgery. The overall incidence of PONV within the first 24 h was also lower in the intervention group compared to the control group ($\chi^2 = 70.5$; $p < 0.001$).

Adverse events

No adverse events (such as fainting, bent needles, stuck needles, bleeding, hematoma or infection) were observed in either group.

Discussion

This RCT evaluated the effectiveness and safety of a single session of immediate postoperative EA on the recovery of gastrointestinal function after LC and found that this intervention significantly shortened the time to first flatus, time to first defecation and time to first normal bowel sounds in these patients. EA also reduced the overall incidence of abdominal pain, abdominal distension and PONV within the first 24 h after surgery and, as such, appears to have effectively promoted the rapid recovery of gastrointestinal

Table 3. Degree/grade of postoperative abdominal distension/pain and nausea/vomiting.

	Intervention group (n=42)	Control group (n=42)	p
Postoperative abdominal distension			
Degree of postoperative abdominal distension at 12 h			0.009
0	39 (93)	30 (71.4)	
I	3 (7)	9 (21.4)	
II	-	3 (7.1)	
III	-	-	
Degree of postoperative abdominal distension at 24 h			0.080
0	40 (95.2)	35 (83.3)	
I	2 (4.8)	7 (16.7)	
II	-	-	
III	-	-	
Postoperative abdominal pain			
Postoperative abdominal pain severity grading at 12 h			0.009
0	31 (73.8)	18 (42.9)	
I	8 (19.0)	20 (47.6)	
II	2 (4.8)	4 (9.5)	
III	1 (2.4)	-	
IV	-	-	
Postoperative abdominal pain severity grading at 24 h			0.017
0	37 (88.1)	28 (66.7)	
I	5 (11.9)	12 (28.6)	
II	-	2 (4.8)	
III	-	-	
IV	-	-	
Postoperative nausea and vomiting			
Nausea and vomiting grading at 12 h			0.013
0	37 (88.1)	27 (64.3)	
I	2 (4.8)	7 (16.7)	
II	4 (9.5)	1 (2.4)	
III	2 (4.8)	2 (4.8)	
IV	2 (4.8)	0 (4.8)	
Nausea and vomiting grading at 24 h			0.030
0	38 (90.5)	26 (61.9)	
I	2 (4.8)	11 (26.2)	
II	2 (4.8)	4 (9.5)	
III	-	1 (2.4)	
IV	-	-	

Data are n (%).

function. Of note, the incidence of POGD in both groups of patients was lower than what has been reported in the literature.^{30,31} Nevertheless, compared with the control group, the degree of abdominal distension at 12 h after surgery, the degree of abdominal pain at 12 and 24 h after surgery and the degree of nausea and vomiting at 12 and 24 h after surgery in the intervention group were all significantly lower, presenting further evidence of the clinical effectiveness of EA in the immediate postoperative phase. Moreover, no adverse events were reported during the trial period, which is reassuring with respect to safety, although ultimately the number of participants was too small to draw definitive conclusions in this regard.

The number of RCTs evaluating acupuncture as a means to enhance recovery of gastrointestinal function after LC is steadily increasing,³² and the advantages of acupuncture in postoperative rehabilitation are becoming more widely recognized. With the booming development of ERAS protocols, LC is increasingly offered as a same-day surgery, with shorter procedures and faster recovery times.^{33,34} Previous studies have highlighted the role of EA in accelerating the recovery of gastrointestinal function after LC, but have often used a large number of needle insertions, multiple treatment sessions and/or lengthy periods of intervention. To address these issues, we designed a pragmatic trial of an immediate postoperative single-session EA intervention to assess its effects on gastrointestinal function under low-intervention conditions, in order to avoid or reduce the discomfort associated with multiple, more intensive and/or long-term acupuncture treatments in postoperative patients.

Acupuncture is a complex intervention and can be highly variable in terms of the anatomical needling targets, selection of which may be important for the achievement of optimal therapeutic effects. POGD is a common complication following abdominal surgery and general anesthesia. Herein we aimed to explore a potentially safer and more effective acupuncture rehabilitation technique by selecting LI4, PC6, LR3 and ST36 as the treatment targets (based on a comprehensive review of the literature and clinical experience), as needling at these locations is believed to have psychological and physiological regulatory effects. Among them, ST36 is widely targeted in the treatment of gastrointestinal disorders including those characterized by abdominal pain and distension. Animal studies have shown that needling at ST36 can promote gastrointestinal neuronal function and improve rhythmic gastrointestinal motility through vagus nerve stimulation while also lowering the levels of vasoactive intestinal peptide (VIP) and nitric oxide (NO) to improve gastrointestinal blood flow, regulate levels of gastric hormones (motilin and somatostatin) and promote recovery of gastrointestinal motility.^{35–39} Research has also indicated that needling at ST36 can reduce the concentration of inflammatory factors, thus inhibiting the occurrence and development of inflammation and reducing gastrointestinal

injury.^{40,41} Acupuncture at LI4 is effective at treating various intestinal disorders, such as diarrhea and abdominal pain, and additionally has sedative and antiemetic effects.^{42–44} It has also been shown that EA at LI4 can increase neuronal excitability and promote gastric dilation.⁴⁵ At the same time, stimulation at LI4 can relax pyloric spasm and have a positive effect on postoperative pain relief.⁴⁶ Acupuncture at LI4 and ST36 in combination is believed to be able to regulate gastrointestinal motility, reduce bloating, relieve pain and promote bowel movements. Clinical research has also demonstrated that acupuncture at ST36 and PC6 in combination can shorten POGD recovery times in patients undergoing LC surgery.⁴⁷ Collectively, acupuncture at these traditional acupuncture point locations are used to treat a variety of disorders involving pain and spasm. PC6 acupuncture has historically been used to treat emotional disorders and facilitate rapid postoperative recovery. Research has shown that stimulation at PC6 can reduce gastric acid secretion and regulate gastrointestinal motility.^{48–50}

During this trial, we subjectively observed that subjects in the intervention group often had more prominent bowel sounds immediately after completion of the EA intervention and some even regained normal bowel function shortly thereafter. In some cases, normal bowel sounds and flatus were restored relatively quickly; however, in most cases, there was a gradual decrease in bowel sounds within 3–6 h of the intervention, followed by a gradual increase until they returned to normal. These observations suggest that EA has a significant and immediate effect on the recovery of postoperative gastrointestinal function. Future studies could explore multiple potential time points for the intervention to optimize its efficacy. Multicenter cohort studies with multifactorial statistical analysis could be conducted to provide a more practical clinical research model for acupuncture research to support implementation and refinement of the treatment approach, which could help identify the optimal acupuncture protocol for various postoperative conditions.

Limitations

This RCT is limited by its small sample size, which could have led to statistical bias in baseline and outcome assessments and is associated with a higher risk of both type I and type II errors. Due to the minimally invasive nature and high technical proficiency of LC, the duration of hospitalization for these patients was generally short, making it challenging to perform longer term clinical observations. Finally, since there is a gradual return of consciousness after general anesthesia, it cannot be guaranteed that patients had no awareness during the 30-min intervention, which began right as anesthesia was being discontinued. Future studies should include a measure of blinding success to address this uncertainty.

Conclusion

An immediate postoperative single-session EA intervention appeared to be effective at promoting the recovery of gastrointestinal function (reflected by return of flatus and normal bowel sounds) and at reducing PONV, abdominal distension and abdominal pain among patients undergoing LC. Overall, the findings of this study are encouraging; however, further research is needed to confirm and extend these findings, as well as to explore the mechanisms underlying the effects of EA as an intervention in this clinical context.

Contributors

QL, WC and W-DS designed the study. QL and HL performed the study. QL conducted the statistical analysis. LT drafted the manuscript. All authors read and approved the final version of the manuscript accepted for publication.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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